

Work through the following exercises while the TA circulates. When you have completed the exercises, spend the rest of the laboratory working on the homework assignment.

Warm up.

Compute the following limits.

1. $\lim_{x \rightarrow 2} e^{-x} \ln(x) \sin(x)$.
2. $\lim_{x \rightarrow 1} \frac{2x-1}{6x^2-7+2}$.
3. $\lim_{x \rightarrow \frac{1}{2}} \frac{2x-1}{6x^2-7+2}$.
4. $\lim_{x \rightarrow 0} \frac{\sqrt{x+1}-1}{x}$.
5. $\lim_{x \rightarrow \infty} \frac{1}{x}$.
6. $\lim_{x \rightarrow \infty} \cos(3x)$.

Exercise 1.

1. Explain the meaning of the expression

$$\lim_{x \rightarrow 2} f(x) = 1.$$

2. Sketch the graph of a function $f(x)$ satisfying the above expression.
3. Explain if such a function can satisfy $f(2) = 1$ or not.

Exercise 2.

Consider the function

$$f(x) = \begin{cases} \frac{x^2-5x+6}{x-2} & \text{if } x < 2, \\ 7 - x^3 & \text{if } x \geq 2. \end{cases}$$

1. Compute $\lim_{x \rightarrow 0} |x|$.
2. Compute $\lim_{x \rightarrow 1} \frac{|x-1|}{x-1}$.
3. Compute $\lim_{x \rightarrow 2^-} f(x)$.
4. Compute $\lim_{x \rightarrow 2^+} f(x)$.
5. Explain the consequence of the last two limits with respect to $\lim_{x \rightarrow 2} f(x)$.